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(#) Covariance :-

covariance is a statistical measure that describes how much two variables change together.

→ Direction (not the strength of the relationship)

(#) correlation :-

correlation is a statistical measure that describe the strength b/w two variables.

→ linear relation
→ direction of the relationship

→ $\boxed{-1 < \rho < 1}$
- close to 1 (+ve correlation)
- close to -1 (-ve correlation)
- close to 0 (No correlation)

→ 0.8 → strong correlation
→ 0.2 → weak correlation

$(-0.5 < \rho < 0.5)$

(No correlation
weak correlation)

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(#) X and y

Pearson's correlation coefficient :-

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(I) $$\rho(x,y) = \frac{cov(x,y)}{std(x) \cdot std(y)} = \frac{cov(x,y)}{\sqrt{var(x)} \cdot \sqrt{var(y)}}$$

(II) $$cov(x,y) = \sum_{i=1}^{N} \frac{(x_i - \mu_x)(y_i - \mu_y)}{N}$$

N: Numbers of values.

(#) Causation :-

Relationship b/w cause & effect

→ it gives the direct connection in which one variables influences the other.

(#) Limitation :-

(1) Outliers :-

(2) it does not measure non linear relationship.

i.e.

correlation & covariance in stock market analysis